Myelodysplastic syndromes

Clonal stem-cell disorders with **ineffective hematopoiesis** → cytopenias, dysplasia, and variable AML risk · blasts **<20%** · management by **lower risk** (treat cytopenias) vs **higher risk** (alter natural history)

Compiled from the Encyclopedia MDS review note and Fellowship Anki deck · WHO 2022 / ICC · IPSS-M (Bernard, NEJM Evid 2022) · Bernard/Papaemmanuil molecular taxonomy (Blood 2024)

DIAGNOSIS

- **Triad:** (1) cytopenia + (2) morphologic **dysplasia in >10%** of a lineage + (3) **clonality** (cytogenetic or molecular). >90% show clonality on karyotype/NGS
- Normal marrow cellularity ≈ **100 – age**. Most MDS is **hypercellular** with ineffective hematopoiesis; a hypoplastic variant overlaps aplastic anemia
- **Dysplasia is not MDS without clonality.** Exclude mimics first: **B12, folate, copper** deficiency, HIV, alcohol, medications, post-chemotherapy marrow recovery
- † Hypoplastic MDS overlaps **aplastic anemia and PNH** — send flow for a PNH clone
- † Consider **VEXAS** (somatic **UBA1**) in older men with inflammatory features, macrocytic anemia and **vacuoles in myeloid and erythroid precursors**

CLASSIFICATION (WHO 2022 / ICC)

Genetically defined	**SF3B1** (favorable, ring sideroblasts) · **isolated del(5q)** (lenalidomide-responsive) · **biallelic TP53** (poor)
Blast categories	**MDS-IB1 = 5–9%** blasts · **MDS-IB2 = 10–19%**
The AML border	At **10–19% blasts with a defining lesion**, **ICC calls it MDS/AML** (managed as AML) while **WHO calls it MDS-IB2**
† del(5q) phenotype	Macrocytic anemia, **normal or high platelets**, hypolobated megakaryocytes, female predominance

RISK STRATIFICATION

IPSS-R (2012)	Cytogenetic risk, marrow blast %, hemoglobin, platelets, **ANC** → 5 groups
IPSS-M (2022)	Keeps blasts, Hgb, platelets, cytogenetics · **drops ANC** · **adds 31 gene mutations** → 6 groups
Mutation weights	**TP53-multihit** highest risk · **SF3B1** favorable · adverse: ASXL1, RUNX1, SRSF2, U2AF1, EZH2, FLT3, KMT2A-PTD

Calculator at mds-risk-model.com. IPSS-M still outperforms a purely molecular-subgroup model, so use it for risk scoring.

Molecular taxonomy (Bernard/Papaemmanuil, Blood 2024). 3,233 patients, 152-gene panel → **16 molecular groups** covering 86%, median OS ranging **0.9–8.2 years**.

Favorable: SF3B1 (14%, most common, OS ~6 yr) · del(5q) · CCUS-like · ZRSR2 · driver-negative. **Monocytic:** bi-TET2 (13%, CMML overlap). **Aggressive:** AML-like (OS ~0.9 yr, 40% 2-yr AML transformation) · TP53-complex · IDH-STAG2 · –7/SETBP1 · EZH2-ASXL1 · BCOR. **DDX41:** high blasts and high transformation **but no excess mortality**.

Take-home: the prognostic weight of **blast %** depends on genetic subtype — but **IPSS-M still outperforms** a groups-only model, so the taxonomy informs biology rather than replacing risk scoring.

TREATMENT PATHWAY

Lower risk — treat cytopenias

Symptomatic anemia
serum EPO <500 + low transfusion burden
→ erythropoiesis-stimulating agent
(epoetin or darbepoetin)

By biomarker
isolated del(5q) → lenalidomide
ring sideroblasts / SF3B1 → luspatercept
ESA-refractory non-del(5q) → imetelstat

Hypoplastic or younger
immunosuppression: ATG ± cyclosporine

Throughout
transfusions; **iron chelation** if chronically transfused; infection prophylaxis

Higher risk — alter natural history

Hypomethylating agent
azacitidine (or decitabine)
† give **≥4–6 cycles before judging response**
AZA-001: OS 24.5 vs 15 months

Transplant-eligible
allogeneic HSCT is the only cure
† timing driven by IPSS-M — do not wait for AML transformation in higher-risk disease

TP53-multihit
poor with all standard therapy
→ prioritize clinical trial or transplant

† After HMA failure
dismal — clinical trial or transplant;
no established standard salvage

BIOMARKER / SETTING → THERAPY

Setting	Therapy	Key trial
Lower risk, EPO <500, low transfusion burden	Erythropoiesis-stimulating agent	—
Lower risk, isolated del(5q)	**Lenalidomide**	MDS-003 / MDS-004
Lower risk, transfusion-dependent, ring sideroblasts / SF3B1	**Luspatercept** — activin receptor II ligand trap, improves late-stage erythropoiesis	MEDALIST; **COMMANDS** (beat epoetin alfa first-line)
Lower risk, ESA-refractory, non-del(5q)	**Imetelstat** — telomerase inhibitor	IMerge (FDA-approved 2024)
Hypoplastic / younger lower risk	Immunosuppression (ATG ± cyclosporine)	—
Higher risk	**Azacitidine** or decitabine	**AZA-001** — OS benefit
Higher risk, transplant-eligible	**Allogeneic HSCT** — the only curative option	—
TP53-multihit	Poor with everything → trial or transplant	—

TRIALS

CALGB 9221 (2002)	The **first RCT** of azacitidine vs supportive care — higher response and **delayed transformation to AML or death**. The disease-modifying signal (Silverman, JCO)
AZA-001 (2009)	Practice-defining — **overall survival benefit** in higher-risk MDS, median OS **24.5 vs 15 months** (Fenaux, Lancet Oncol)
† Decitabine (D-0007, EORTC 06011)	Improved response and delayed AML progression but **no OS benefit**. Caveat: decitabine was capped at 6 cycles (median 4) while azacitidine continued to progression (median 9). Practically, **azacitidine is the HMA with an OS benefit**
MDS-003 / MDS-004	Lenalidomide in transfusion-dependent lower-risk **del(5q)** — ~67% transfusion independence (List, NEJM 2006)
MEDALIST (2020)	Luspatercept after ESA failure, ring sideroblasts / SF3B1 — transfusion independence **38% vs 13%**
COMMANDS (2023)	Luspatercept **beat epoetin alfa first-line** — ~59% vs ~31%
IMerge (2024)	Imetelstat in ESA-refractory non-del(5q) — 8-week transfusion independence **40% vs 15%**
† BMT CTN 1102	Donor vs no-donor in higher-risk MDS age 50–75 — 3-yr OS **47.9% vs 26.6%**. The evidence for transplanting older patients (Nakamura, JCO 2021)
† VERONA (2025)	Venetoclax + azacitidine in higher-risk MDS **missed its OS endpoint** (22.2 vs 21.7 mo, HR 0.91) despite higher response rates — **ven/aza is standard in AML but not MDS**